SESSION 03

Bank Branch Security Lab

Phase 3 · Endpoints: a domain-joined workstation · October 9, 2026

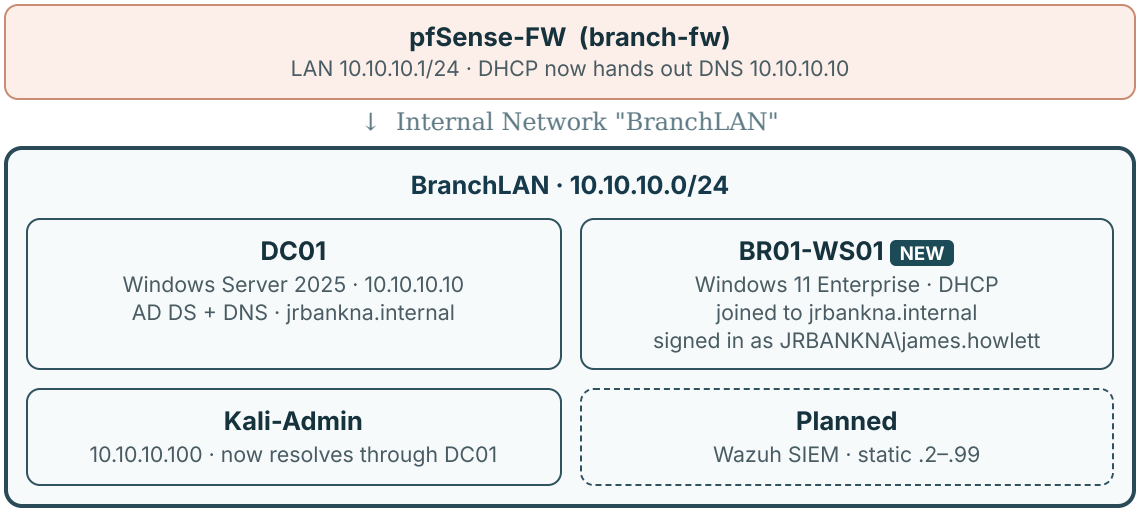
Goal	Put the first branch workstation on the domain: a Windows 11 PC that finds the domain controller through DNS and lets a teller sign in with domain credentials.
Outcome	Branch DHCP now hands out DC01 as the DNS server, BR01-WS01 was built and joined to jrbankna.internal, and a teller account signed in with domain credentials.
Systems built	BR01-WS01 (Windows 11 Enterprise Evaluation) · DHCP DNS change on pfSense-FW
Branch network	Unchanged: 10.10.10.0/24 · DHCP clients now use DC01 (10.10.10.10) for DNS
Skills	DHCP options, DNS and AD integration, Windows 11 deployment (UEFI, TPM 2.0, Secure Boot), domain join, least privilege, endpoint troubleshooting

Overview

This phase connected the first endpoint to the branch's identity system. I changed the pfSense DHCP server so every branch client receives DC01 (10.10.10.10) as its DNS server and jrbankna.internal as its domain, then confirmed DC01 answers for the domain and still resolves outside names through the firewall. I built a Windows 11 Enterprise workstation, BR01-WS01, with the EFI, TPM 2.0 and Secure Boot settings Windows 11 requires, installed it with a local account instead of a Microsoft account, and renamed and joined it to the domain in one step using the separate admin account. After the restart, a standard teller account signed in with domain credentials.

Architecture

The perimeter and the domain controller are unchanged. What changed is how clients find things: DHCP now tells every branch device to use DC01 for DNS, and the first domain-joined workstation sits on the LAN.



DHCP change (pfSense › Services › DHCP Server › LAN)

Field	Before	After
DNS Servers	Blank, so clients received pfSense (10.10.10.1)	10.10.10.10 (DC01 only)
Domain Name	Blank (home.arpa)	jrbankna.internal

Accounts used on BR01-WS01

Account	Type	Used for
Local account created at setup	Local; this PC only	Fallback sign-in if the domain can't be reached; not for daily use
`JRBANKNA\adm.jryounger`	Domain admin	Authorizing the domain join only
`JRBANKNA\james.howlett`	Standard domain user	Everyday sign-in, like a teller starting a shift

What I built

1. In the pfSense DHCP server, set DNS Server to 10.10.10.10 and Domain Name to jrbankna.internal, confirmed the pool still excludes .10, and applied.
2. Renewed Kali's lease and verified from `/etc/resolv.conf` that it now uses DC01; `nslookup dc01.jrbankna.internal` returned 10.10.10.10 and an outside name still resolved through the forwarder.
3. Created the BR01-WS01 VM: Windows 11 (64-bit), 8 GB RAM, 2 vCPU, EFI on, 80 GB dynamic disk, on BranchLAN.
4. Confirmed TPM 2.0 and Secure Boot in Settings, then installed Windows 11 Enterprise Evaluation as a clean install.
5. At setup, chose Sign-in options › Domain join instead, creating a local account rather than tying the PC to a Microsoft account.
6. Set the computer name to BR01-WS01 and joined it to jrbankna.internal in one change (`sysdm.cpl`), authorized with adm.jryounger, and restarted.
7. Signed in as the teller james.howlett: a standard domain account on a domain-joined workstation.
8. Planned a single shared snapshot of all four VMs, taken after a graceful shutdown.

Design decisions

Decision	Rationale
DC01 as the only DNS server for clients	Domain members find domain controllers through DNS records only DC01 holds. A public backup such as 8.8.8.8 would sometimes answer "that domain doesn't exist," causing random login failures.
Change DHCP before building the workstation	The join depends on DNS. Fixing and testing name resolution first, from a client that already existed, isolated the change before adding a new machine.
Name it BR01-WS01	Branch, device type and number. A location-based name stays accurate and sorts by branch in logs and the SIEM.
Local account at setup, no Microsoft account	A bank PC's identity belongs to the organization. A personal cloud account would mix personal and corporate data outside central control.
Rename and join in one step	One restart, and the computer object is created in AD with its final name instead of a random DESKTOP-XXXXXX name that would need cleanup.
Join with adm.jryounger, sign in as a teller	Admin credentials are typed once, only to authorize the join. Daily work happens on an account with no admin rights.
EFI, TPM 2.0 and Secure Boot	Required by Windows 11, and the foundation for BitLocker and verified boot that banks rely on for laptops.

Verification

Check	How	Result
DHCP hands out DC01 as DNS	`cat /etc/resolv.conf` (Kali)	Pass · nameserver 10.10.10.10
DHCP hands out the domain name	`cat /etc/resolv.conf`	Pass · search jrbankna.internal
DC01 answers for the domain	`nslookup dc01.jrbankna.internal`	Pass · 10.10.10.10
Outside names via the forwarder	`nslookup google.com`	Pass · resolved via 10.10.10.10
VM hardware as planned	VirtualBox wizard summary	Pass · 8 GB, 2 vCPU, EFI, 80 GB
Windows 11 installed	Desktop watermark	Pass · Enterprise Evaluation
Joined to the domain	`sysdm.cpl` › Change	Pass · Welcome to jrbankna.internal
Teller signs in with domain account	Other user › JRBANKNA\james.howlett	Pass · desktop loads

Troubleshooting

VM opened the firmware menu instead of Windows Setup

Cause	Windows install media on a UEFI machine waits only a few seconds at "Press any key to boot." With no key pressed and an empty disk, the firmware stopped at its menu.
Fix	Boot Manager › UEFI optical drive, then tapped Spacebar immediately until Setup loaded. Firmware menus are keyboard-driven.

Windows 11 setup pushed a Microsoft account

Cause	Windows 11 setup defaults to cloud sign-in.
Fix	Chose Sign-in options › Domain join instead, which creates a local account and leaves the domain join for after setup.

Workstation clock three hours behind

Cause	A three-hour offset pointed to the time zone (Windows setup's default Pacific), not a wrong clock. Kerberos compares UTC, so the sign-in still worked.
Fix	Set Eastern in Settings, or `tzutil /s "Eastern Standard Time"`. Log timestamps must line up across systems before the SIEM goes in.